Jonathan Barnes | Data Scientist

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PROFESSIONAL EXPERIENCE

University of Minnesota, Minneapolis, MN

Graduate Research Assistant | Center for Public Health Systems *(May 2025 - Present)*

- Owned an end-to-end federal health-funding data system spanning ingestion, normalization, storage, and visualization.
- Engineered a scalable ELT pipeline in Python (Polars, DuckDB, SQL) processing 3M+ transactions from USASpending.gov.
- Implemented custom entity-resolution and award-matching logic to reconcile prime and sub-awards, parent–child recipients, and government health entities, while optimizing storage via Parquet (77%+ size reduction).
- Delivered production Tableau dashboards powered by Hyper extracts to visualize and communicate funding trends.

Graduate Teaching Assistant - In-Person and Virtual | Division of Biostatistics and Health Data Science *(Aug 2024 - May 2025)*

- Taught weekly lab sessions in R and SAS to develop student proficiency in statistical analysis and programming.
- Evaluated student projects and quizzes, delivering actionable feedback that accelerated learning outcomes.

Graduate Research Assistant | Medical School *(Oct 2023 - May 2024)*

- Classified plasma biomarker profiles from NIA funded study into cutoff-based phenotypes, driving downstream modeling.
- Modeled time to cognitive decline with Cox proportional hazards regression, assessing predictive power for dementia.
- Programmed differential gene expression analysis stratified by grouping and recorded cognitive decline.

University of St. Thomas, St. Paul and Minneapolis, MN

ITS Support Technician Intern & Senior Student Technician | ITS Support Services *(Aug 2021 - Aug 2023)*

- Resolved or consulted on at least 4,500 departmental tickets over two years.
- Engineered and developed a Python application vital to ITS service desk function and operational efficiency.
- Identified and resolved process gaps, ensuring efficient workflows by proactively addressing vulnerabilities.
- Mentored numerous junior students, to equip them to independently handle high-level operations.
- Implemented digital signage across service desks and various campuses, to provide accessible information access.
- Imaged and administered software on over 1,500 summer replacement computers, over double the usual volume.
- Assembled in-house virtual lab to reduce existing AWS EC2 cloud instances and reduce operational costs.
- Authored technical documentation and procedural guides used by The Tech Desk, ITS staff, students, and faculty.

EDUCATION

University of Minnesota School of Public Health, Minneapolis, MN

Master of Public Health - Public Health Data Science, GPA 3.73/4.00 *(Expected Jan 2026)*

- APEx - Pre-Statistics Modules for MPH Students | ILE – Evaluation of US Federal Spending and Systems.

University of St. Thomas, St. Paul, MN

Bachelor of Science - Data Analytics, Biology Domain, Completed concurrently with full-time employment.

- Minors - Biology (Biochemistry & Computational Biology courses), Applied Statistics.
- Capstone - Development of Obesity and Nutritional Trends Over Time in the United States Using NHANES.

AWARDS & RECOGNITION

- **President's Student Leadership & Service Award Nominee**, University of Minnesota (2025)
- **ITS Outstanding Student Award**, University of St. Thomas (2023)

LEADERSHIP AND VOLUNTEERING

Biostatistics Community Outreach & Engagement (BCOE), *Member (Sep 2023 - Present)*

School of Public Health Student Senate (SPHSS), *BSHD Representative (Oct 2023 - May 2025)*

- *Committees – Health Science Student Consultive Committee, Advocacy, Constitution Revision, Lounge Modernization.*

SKILLS & RELEVANT METHODS

Languages: R, Python, SQL, Bash.

Tools: Quarto, Shiny, APIs, Git, Linux, PowerShell, ArcGIS, Tableau, CI/CD, Azure, Docker, Arrow.

Competencies: Data Visualization, ETL Pipelines, Data Communication, Geospatial Visualization, Feature Engineering, Predictive Modeling, Dimensionality Reduction, Clustering, Classification, Longitudinal Data Analysis, AI, HIPAA, PHI, Informatics, Relational Databases, Electronic Health Records (EHR), SNOMED, ICD-10, Columnar Data, System Design, Consulting, Experimental Design.

Statistical and Machine Learning Methods: Regression, Random Forest, Gradient Boosting, Principal Component Analysis (PCA), Multiple Correspondence Analysis (MCA), Generalized Estimating Equations (GEE), Generalized Mixed Effects Models (GLMM), Regularized Regression, Cox Proportional Hazards, Structural Equation Models (SEM), Natural Language Processing (NLP).

PROJECTS

Mapping Misery – Predicting Poor Physical and Mental Health Days in the U.S.

- Developed predictive models for poor physical and mental health days in 3,000+ U.S. counties using R.
- Engineered clusters based on 10 years of trends across 15 metrics using Gaussian Mixture Models (GMM) by transforming variables into 8 level categories with NAs for robust Multiple Correspondence Analysis (MCA).
- Achieved an R^2 of 0.844 (RMSE: 0.263) for physical health days with Elastic Net Regression.

  Methods & Tools: Data Pipelines, PCA, MCA, GMM Clustering, Feature Engineering, Longitudinal Data Analysis, Elastic Net Regression, Predictive Modeling, R, Geospatial Visualization, GIS, Data Visualization, Data Communication.

Tech Desk Tools Python

- Built a GUI-based desktop application for the University of St. Thomas Tech Desk to streamline common support tasks such as ticket creation, password resets, system diagnostics, and resource access in use continuously since February 2023.
- Redesigned and modernized legacy AutoHotKey application in Python, improving functionality and future scalability.
- Integrated role-specific admin features for senior student technicians based on assigned Azure roles.
- Deployed the application to all staff and student workers at the University of St. Thomas's Tech Desk.

  Methods & Tools: Python, PowerShell, Bash, Dashboards, Data Communication, GUI Development, Active Directory, Azure, Technical Writing, Packaging and Deployment.

Public Health System Federal Funding Dashboard (Capstone, Grant Supported)

- Designed and owned an end-to-end data system integrating over 3 million U.S. Department of Health and Human Services funding transactions from USASpending.Gov.
- Implemented ingestion, normalization, enrichment, and reconciliation logic to align awards with health departments.
- Engineered a columnar storage and query architecture using Apache Parquet and DuckDB, reducing data size.
- Built a GUI-based application to manage pipeline execution and data refreshes, enabling human-in-the-loop operation.
- Generated Tableau Hyper extracts that directly feed interactive dashboards in Tableau for analysis.

  Methods & Tools: Python, Polars, SQL, DuckDB, Apache Parquet, Tableau, Hyper, GUI Development, Data Pipelines, Data Validation, Data Engineering, Data Visualization, Data Communication.

Data Inspection Dashboard | https://jonathanbarnes.shinyapps.io/DescriptiveApp/

- Developed an interactive tool that allows users to conduct baseline Exploratory Data Analyses (EDAs) without requiring R knowledge and accepts common dataset formats in public health.
- Automates and facilitates data cleaning, NA handling, imputation, and produces a data table with inline distribution plots and summary statistics.

  Methods & Tools: R, Shiny, Data Pipelines, Dashboards, Data Visualization, Data Communication, JavaScript.

Applied Biostatistics Introduction Modules (Capstone, Grant Supported)

- Designed modular teaching materials introducing probability, inference, and regression through applied public health examples at the request of Public Health Administration & Policy program director
- Developed intuition-focused explanations to support students with apprehension toward statistics.
- Addressed recurring conceptual gaps identified by faculty and me from teaching.
- Emphasized ethical interpretation and justification of statistical choices in applied public health contexts.

  Methods & Tools: Biostatistics, Statistical Reasoning, Probability, Curriculum Design, Technical Writing, R, Data Communication, Public Health Education.

Biostatistics Reference Guide

- Provides step-by-step instructions and R code for linear regression, logistic regression, Poisson regression, Cox proportional hazards models, survival analysis, ordinal regression, and log-binomial regression.

  Methods & Tools: R, Regression Analysis, Survival Analysis, Logistic Regression, Poisson Regression, Ordinal Regression, Log-binomial Regression, Tidyverse, ggplot2, Technical Writing.

Predicting Child Mortality and Cesarean Delivery Outcomes with Generalized Linear Mixed Models.

- Built Generalized Linear Mixed Models (GLMM) to investigate factors affecting child mortality and cesarean delivery.
- Determined a 1,124.5% increase in the odds of cesarean delivery for hospital births (95% CI: 618.7% - 1,986.3%, $p < 0.001$) — a result consistent with logical expectations.

  Skills & Tools: GLMM, R, Data Cleaning, Data Pipelines, Data Visualization, Feature Engineering, Collaboration.